

Aditya Sagar Sharma

adityasagar9991@gmail.com — +91 9709303105 — github.com/adityass2004 —
linkedin.com/in/aditya-sagar-sharma-1955a7288/ — adityass.vercel.app

Education

SRM Institute of Science and Technology

Chennai, India

B.Tech in Computer Science and Engineering **CGPA: 8.97 / 10.0**

Aug 2023 – May 2027

Relevant Coursework: Data Structures & Algorithms, Design & Analysis of Algorithms, Operating Systems, DBMS, Computer Networks, Artificial Intelligence, Machine Learning, NLP, Discrete Mathematics, Formal Languages & Automata

Research (Ongoing)

Damage-Guided Adaptive Recovery for Efficient Neural Network Pruning

Under Review

Dept. of Computing Technologies, SRMIST KTR

Proposed a **damage-aware pruning framework** that addresses irreversible performance degradation in deep learning model compression. The method introduces a *composite importance metric* combining weight magnitude, gradient sensitivity, and activation frequency to identify redundant parameters. A *per-layer damage metric* (mean squared deviation between original and pruned activations) pinpoints the most severely affected layers, after which *selective weight restoration* and targeted retraining recover representational capacity. Experiments on **CIFAR-10, CIFAR-100, and FashionMNIST** (VGGSmall) demonstrate **70 % parameter reduction** and **≈60 % FLOP reduction** while maintaining or surpassing baseline accuracy at up to 70 % sparsity.

Keywords: Neural Network Pruning, Model Compression, Deep Learning, Sparsity, CIFAR-10/100, PyTorch

Technical Skills

Languages	C++, JavaScript (ES6+), TypeScript, Python, Dart
Backend	Node.js, Express.js, Next.js API Routes, REST APIs, JWT, PostgreSQL, MySQL, MongoDB
Frontend	React.js, Next.js, Tailwind CSS, HTML/CSS
AI / ML	RAG, LLM APIs, Computer Vision, NLP, Neural Network Pruning, PyTorch, Jupyter Notebook
Tools	Git, GitHub, Docker, Firebase, Vercel, Netlify, AWS (Basic), Arduino

Experience

Cognifyz Technologies

Remote

Web Development Intern

Jul 2025 – Aug 2025

- Optimized Node.js and Express.js backend services, achieving a **30 % reduction** in average task completion time through query optimization and middleware restructuring.
- Delivered **5+ production-ready** backend features covering API design, database operations, and input validation across Agile sprint cycles.
- Maintained **95%+ code quality score** as measured by internal review metrics.

Sandeshkhali Maa Saroda Women & Rural Welfare Society

West Bengal

Community Connect Volunteer

Jun 2025 – Jul 2025

- Conducted **40+ hours** of digital literacy workshops enabling **50+ rural women** to independently navigate digital government services.

Projects

PDF Chat – AI-Powered Document Analysis

Dec 2025 [GitHub]

- Built an end-to-end **RAG pipeline** enabling semantic Q&A over PDF documents using LLM APIs and vector embeddings; integrated Vision Models for image-rich PDFs.
- Stack: *Python, LLM APIs, RAG architecture, PDF parsing libraries.*

Social Media Platform – Backend-First Full-Stack

Sep 2025 [GitHub]

- Designed scalable backend supporting profiles, posts, likes, comments, and follow relationships with JWT auth across dual **PostgreSQL + MongoDB** architecture.
- Stack: *Node.js, Express.js, PostgreSQL, MongoDB, React.js.*

Track-It – Academic Productivity Platform

Aug 2025 [GitHub]

- Full-stack system unifying attendance, placement notices, and student records with role-based access on **Next.js API routes** backed by optimized MongoDB schemas.

Airline Reservation System

Nov 2024 [GitHub]

- Complete airline management system with flight booking, seat allocation, and admin dashboard; relational **MySQL** schema with transactional integrity.
- Stack: *Next.js, MySQL, Tailwind CSS, REST APIs*.

StareWare – Real-Time In-Browser Gaze Detection

Jul 2025 [GitHub]

- Real-time **face detection, eye tracking, and presence monitoring** in-browser via webcam using browser-native CV APIs – zero backend, deployed as standalone app.

Certifications

- Oracle Cloud Infrastructure (OCI) 2025 Certified Foundations Associate
- Generative AI Foundations – Google Cloud / AWS
- Networking Basics – Cisco Networking Academy

Achievements

- Solved **65+ DSA problems** on LeetCode (graphs, DP, trees)
- Participated in **HackRx 6.0** hackathon
- Live portfolio deployed – **10+ projects** showcased